Basics 29: Linearization of Magnetohydrodynamic Equations

IV - Magnetohydrodynamics

Equilibrium-plus-perturbation expansions, displacement variables, normal modes, and eigenvalue structure

Textbook draft, version 1.0. Linearization converts nonlinear MHD into a wave and stability problem around a chosen equilibrium. This chapter derives the first-order equations, introduces the Lagrangian displacement, and explains how a time-dependent PDE becomes an eigenvalue problem for mode frequency and structure. This chapter is designed for a reader with no prior plasma-physics training. It distinguishes exact identities from physical approximations and connects the central equations to later simulation modules.

Contents

1 Learning objectives

After completing this note, the reader should be able to:

- Perform a controlled first-order perturbation expansion.
- Derive linearized continuity, induction, pressure, and momentum equations.
- Relate Eulerian perturbations to a Lagrangian displacement.
- Apply normal-mode substitution to obtain an eigenproblem.
- Distinguish equilibrium constraints from perturbation equations.
- State the validity limits of linear theory.

2 Prerequisites and reading strategy

- Basics 5-8, 24, and 26.

On a first reading, emphasize physical meaning and dimensional checks. On a second reading, reproduce the derivations. Attempt the computational laboratory only after the limiting cases are understood.

3 Equilibrium and perturbation ordering

Write every field as equilibrium plus a small perturbation:

$$Q(\mathbf{x},t) = Q_0(\mathbf{x}) + \epsilon Q_1(\mathbf{x},t) + O(\epsilon^2), \qquad \epsilon \ll 1. \tag{3.1}$$

For a static equilibrium $\mathbf{u}_0 = 0$,

$$\nabla p_0 = \mathbf{J}_0 \times \mathbf{B}_0, \qquad \nabla \cdot \mathbf{B}_0 = 0. \tag{3.2}$$

Substitute expansions into the full equations, cancel the $O(1)$ equilibrium balance, retain $O(\epsilon)$ terms, and discard products such as $\rho_1\mathbf{u}_1$.

4 Linearized continuity and pressure

Mass conservation gives

$$\partial_t\rho_1 + \nabla\cdot(\rho_0\mathbf{u}_1) = 0. \tag{4.1}$$

The adiabatic pressure equation yields

$$\partial_t p_1 + \mathbf{u}_1\cdot\nabla p_0 + \gamma p_0\nabla\cdot\mathbf{u}_1 = 0. \tag{4.2}$$

Equilibrium gradients therefore couple displacement to compressibility even when perturbation amplitudes are small.

5 Linearized induction equation

From

$$\partial_t\mathbf{B} = \nabla\times(\mathbf{u}\times\mathbf{B}), \tag{5.1}$$

the first-order equation is

$$\partial_t\mathbf{B}_1 = \nabla\times(\mathbf{u}_1\times\mathbf{B}_0), \qquad \nabla\cdot\mathbf{B}_1 = 0. \tag{5.2}$$

The term $\mathbf{u}_1\times\mathbf{B}_1$ is second order and is omitted.

6 Linearized momentum equation

For a static equilibrium,

$$\rho_0\partial_t\mathbf{u}_1 = -\nabla p_1 + \mathbf{J}_1\times\mathbf{B}_0 + \mathbf{J}_0\times\mathbf{B}_1, \tag{6.1}$$

where

$$\mathbf{J}_1 = \frac{1}{\mu_0}\nabla\times\mathbf{B}_1. \tag{6.2}$$

Both perturbed current acting on the equilibrium field and equilibrium current acting on the perturbed field are first order.

7 Lagrangian displacement

Introduce displacement $\boldsymbol{\xi}$ by

$$\mathbf{u}_1 = \partial_t \boldsymbol{\xi}. \tag{7.1}$$

Integrating the linearized continuity, pressure, and induction equations for perturbations initially consistent with the displacement gives

$$\rho_1 = -\nabla \cdot (\rho_0 \boldsymbol{\xi}), \tag{7.2}$$

$$p_1 = -\boldsymbol{\xi} \cdot \nabla p_0 - \gamma p_0 \nabla \cdot \boldsymbol{\xi}, \tag{7.3}$$

$$\mathbf{B}_1 = \nabla \times (\boldsymbol{\xi} \times \mathbf{B}_0). \tag{7.4}$$

These relations express all perturbed fields in terms of one geometric variable.

8 Normal modes and the force operator

Substitute $\boldsymbol{\xi}(\mathbf{x}, t) = \hat{\boldsymbol{\xi}}(\mathbf{x}) e^{-i\omega t}$. The momentum equation becomes

$$-\omega^2 \rho_0 \hat{\boldsymbol{\xi}} = \mathcal{F}[\hat{\boldsymbol{\xi}}], \tag{8.1}$$

or

$$\mathcal{A} \hat{\boldsymbol{\xi}} = \omega^2 \mathcal{M} \hat{\boldsymbol{\xi}}. \tag{8.2}$$

Under ideal boundary conditions, the force operator has a self-adjoint energy structure. Positive ω^2 gives oscillation; negative ω^2 corresponds to $\omega = i\gamma$ and exponential instability.

9 Eulerian and Lagrangian perturbations

An Eulerian perturbation compares fields at a fixed position. A Lagrangian perturbation follows a displaced fluid element:

$$\Delta Q = \delta Q + \boldsymbol{\xi} \cdot \nabla Q_0. \tag{9.1}$$

For density and pressure,

$$\Delta \rho = -\rho_0 \nabla \cdot \boldsymbol{\xi}, \qquad \Delta p = -\gamma p_0 \nabla \cdot \boldsymbol{\xi}. \tag{9.2}$$

Confusing δQ and ΔQ is a common source of sign and gradient errors.

10 Validity and nonlinear continuation

Linear theory is valid while perturbations remain small relative to equilibrium scales and do not significantly modify particle distributions or equilibrium profiles. It predicts frequencies, mode shapes, and initial growth rates, but not nonlinear saturation. In this project, the Module 4 eigenproblem is linear; Modules 5-8 later supply energetic-particle drive, transport, and feedback.

11 Computational laboratory

Use symbolic algebra to linearize a one-dimensional nonlinear conservation law and the uniform-field ideal-MHD equations. Construct the shear-Alfvén matrix eigenproblem, verify eigenvector normalization and residuals, and compare time-domain oscillation frequency with the eigenvalue.

12 Summary

- Linearization is an ordered expansion about an equilibrium, not simply deletion of nonlinear-looking terms.
- Continuity, pressure, induction, and momentum perturbations are coupled.
- The Lagrangian displacement expresses the perturbation system through one primary variable.
- Normal-mode substitution produces a generalized eigenvalue problem.
- Linear theory determines initial waves and stability but not nonlinear saturation.

Symbols and acronyms used in this document

Symbol / acronym	Name	Meaning in this document
Q_0, Q_1	equilibrium and perturbation	Zeroth- and first-order fields.
ϵ	perturbation parameter	Formal amplitude ordering.
$\boldsymbol{\xi}$	Lagrangian displacement	Fluid-element displacement.
δQ	Eulerian perturbation	Change at fixed position.
ΔQ	Lagrangian perturbation	Change following a fluid element.

Symbol / acronym	Name	Meaning in this document
$\mathcal{F}$	linear force operator	Restoring or destabilizing MHD force.

13 Exercises

1. Linearize the product $\rho\mathbf{u}$ about a static equilibrium.
2. Derive the displacement expression for ρ_1.
3. Derive $\mathbf{B}_1 = \nabla \times (\boldsymbol{\xi} \times \mathbf{B}_0)$.
4. Explain why $\mathbf{J}_0 \times \mathbf{B}_1$ must be retained.
5. Interpret positive, zero, and negative values of ω^2.

14 References

References

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